



Stress waves in thick porous graphene-reinforced cylinders under thermal gradient environments

Rasool Moradi-Dastjerdi, Kamran Behdinan *

Advanced Research Laboratory for Multifunctional Lightweight Structures (ARL-MLS), Department of Mechanical & Industrial Engineering, University of Toronto, Toronto, Canada

ARTICLE INFO

Article history:

Received 23 October 2020
Received in revised form 8 December 2020
Accepted 23 December 2020
Available online 7 January 2021
Communicated by Jang-Kyo Kim

Keywords:

Stress wave propagation
Graphene-reinforced cylinders
Embedding porosity
Structural damping
Axisymmetric model
Thermal gradient load

ABSTRACT

In this paper, the propagation of stress waves in thick porous graphene-reinforced cylinders (TPGRCs) is investigated using a meshless solution based on an axisymmetric model and moving least squares (MLSs) interpolation functions. The cylinders are assumed under a steady state thermal gradient effect along with an initial deflection response obtained from the static solution of the same cylinder subjected to a static internal pressure. To improve the design of such TPGRCs, functionally graded (FG) profiles are considered for the distribution of graphene nanofillers along their radial direction. Moreover, pores are non-uniformly embedded inside the cylinders to decrease the weight of TPGRCs. The material properties of porous and nanocomposite materials are calculated as functions of temperature using a closed cell model and a modified Halpin-Tsai (HS) technique, respectively. Considering structural damping for the system, the influences of embedding porosity, the temperature of each surface, reinforcing with graphene and the thickness of cylinder on the stress wave responses of TPGRCs have been investigated. The results reveal that the increase of porosity volume inside TPGRCs significantly increases the amplitudes and stationary time of deflections; however, it reduces the values of stationary deflections and stresses, along with the reduction of propagation speeds.

© 2020 Elsevier Masson SAS. All rights reserved.

1. Introduction

Increasing demands for multifunctional lightweight structures in industries have being speeded up developments in the technologies of manufacturing new lightweight materials with adjustable thermo-mechanical properties. As lightweight materials, traditional composites have seen greater attentions over that past three decades [1]. However, after the inventions of nanofillers and FG materials, traditional composites are being replaced by nanocomposites or FG materials. Nanofillers usually enjoy astonishing mechanical, thermal and electrical properties where the use of small amounts of them in a matrix makes great differences in the material properties of that matrix/nanofiller material without a notable change in the total weight [2–4]. Graphene nanosheets with a tightly packed structure and a thickness equal to one single atom is one of the most important nanofillers which has

exceptional electrical/thermal conductivity along with a remarkable high Young's modulus [5,6]. Furthermore, FG materials which concurrently have the benefits of metals and ceramics, are being used in high-tech industries. These materials are usually made of a ceramic/metal mixture with adjustable distribution of material contents. Also, porous materials like polymeric foams are another type of lightweight materials [7–9]. To have all the advantages of nanocomposite, porous and FG materials, one can involve all these three materials and propose FG porous nanocomposite materials/structures [10–13]. In these multifunctional lightweight materials, the distribution of either nanofillers or pores can be considered in FG patterns along a certain direction(s) [14–16]. Such advanced materials have an extensive potential of use for different engineering applications including aerostructures [17], energy storage [18], and gas barrier [19,20].

After the observation of a great enhancement on the thermal and mechanical properties of polymers enhanced with either carbon nanotube (CNTs) [21,22] or graphene nanosheets [23–25], thermo-mechanical behaviors of engineering structures which involved such nanocomposite materials have attracted substantial attentions from researchers [26–28]. In addition, the impacts of

* Corresponding author.

E-mail addresses: moradi@mie.utoronto.ca (R. Moradi-Dastjerdi), behdinan@mie.utoronto.ca (K. Behdinan).

these two important nano-reinforcements on the transient heat conductions of FG nanocomposite cylinders [29] and cylindrical panels [30] were investigated. Using 3D piezo-elastic theory and an analytical method, Alibeigloo [31] studied the thermo-electro-mechanical stresses of CNT-reinforced cylindrical panels located between two piezoelectric faces. Shen et al. [32] developed perturbation technique to study nonlinear deflections caused by the effect of thermal environments for thin-walled FG nanocomposite cylindrical panels reinforced with graphene nanosheets. Yang et al. [33] developed an analytical solution based on 3D elasticity theory to present the thermoelastic stresses of circular disks and plates made of FG polymer/graphene platelets. Using a quasi-steady modified piston theory to model the aerodynamic pressure, Asadi [34] studied flutter phenomenon for FG CNT-reinforced thin-walled cylinders under aerodynamic loads and considering a thermal environment. Sobhaniragh et al. [35] explored stresses caused by steady-state thermo-mechanical loads for thin ceramic cylindrical shells reinforced with the agglomerations of randomly oriented CNTs using generalized differential quadrature (DQ) technique. Pourasghar et al. [36] studied the thermo-elastic stress responses of CNT-reinforced cylindrical panel and cylinders using 3D elasticity theory with appropriate displacement and temperature functions capable of the satisfaction of boundary conditions. The natural frequencies of FG graphene-reinforced circular disks with an eccentric hole and activated by two piezoelectric faces were presented by developing a DQ technique in [37]. Liu and Behdinan [38] developed a novel analytical technique with the aim of reducing computational costs to study heat transfer analysis in fiber-reinforced plastic pulleys in belt drive systems. Using a third order theory of plates, nonlinear natural frequencies of FG graphene platelet-reinforced plates were studied using numerical techniques in [39,40]. Considering thermal gradient effects, natural frequencies along with the displacement and stress fields of axisymmetric FG cylinders reinforced with graphene nanosheets [41] and wavy CNTs [42] were investigated using a meshless solution.

In a framework of dynamic study, Song et al. [43] considered FG polymer/graphene platelets plates and studied the vibrations of such plates under pulse excitations using Navier technique. By applying harmonic excitations with loading frequency equal to the natural frequencies of system, Moradi-Dastjerdi et al. [44] investigated resonance phenomenon in axisymmetric cylinders reinforced with wavy CNTs. For the same nanocomposite cylinders, they also investigated the effect of thermal gradient and half sine impact loads on the propagation of stress waves using a meshless solution [45]. By considering viscoelastic foundations, nonlinear vibrations in FG polymer/graphene platelets plates were investigated using Runge-Kutta method [46], and a novel analytical approach [47]. In addition, the propagations of elastic waves inside cylinders [48] and rotating cylindrical shells [49] made of FG polymer/graphene were studied using finite difference technique and Rayleigh-Ritz approach, respectively. Babaei et al. [50] conducted an analytical study on the frequency and transient responses of FG porous metallic disks and panels. Their formulations were developed based on 3D elasticity theory.

Damped harmonic vibrations along with stability resistances and frequencies of sandwich plates which involved FG graphene nanosheets/polymer, FG porous core and piezoelectric layers were studied in [51–53]. Considering FG distributions for the graphene platelets and pores, different mechanical analyses of porous nanocomposite curved beams including vibrations, deflections and buckling stability were presented in [54]. Nguyen et al. [55] presented static and vibration behaviors of flat plates made of such porous nanocomposite materials integrated between piezoceramics by developing an Finite element method. For cylindrical shells with concurrently embedded pores and graphene platelets in FG distribution patterns, natural frequency analysis was conducted us-

ing Galerkin's method in [56]. In addition, the buckling analysis of such cylinders with considering rotating effects were reported in [57]. Furthermore, recently, some other porous beams [58,59], plates [60] and shells [61] reinforced with graphene nanofillers were considered and their mechanical analyses were reported. While the study of stress waves or considering thermal gradient conditions has not properly been investigated yet.

In aerostructures, fuel efficiency and safety are two important design goals. Therefore, the use of advanced multifunctional lightweight materials and precise modeling of these structures are crucial. Moreover, such structures are usually subjected to thermal gradient environment and time-dependent mechanical loads. These types of loadings can be observed in nuclear reactors and different types of engines as well. Hence, it can be sensed that such structures need sophisticated analyses especially when they are made of advanced materials. Cylindrical structures, as widely used structures, can be considered as a case study to characterize their thermoelastic dynamic responses. However, as observed in the literature review, the propagation of stress waves in nanocomposite cylinders has been rarely investigated before. Furthermore, such dynamic analysis for porous graphene nanocomposite cylinders, as multifunctional lightweight structures, under thermal gradient loads has not been reported yet. Therefore, as the first endeavour, this work aims to address this gap by presenting the analysis of stress waves propagation in thick axisymmetric nanocomposite cylinders with FG embedded graphene nanosheets and pores under a steady state thermal gradient loads and an initial condition for dynamic responses. With this aim, an axisymmetric meshless solution along with HS technique are employed. Considering structural damping for the system, the influences of porosity, temperature, graphene and the thickness of cylinder on the propagation of stress waves in such TPGRs have been investigated.

2. Modeling of axisymmetric TPGRs

The TPGRs considered in this paper are assumed to be made of a polymeric material with concurrently embedding pores and randomly oriented graphene nanosheets. Along the radial direction of TPGRs, FG profiles of graphene nanosheet and porosity distributions are assumed. The inner and outer surfaces of TPGRs are also assumed under different temperatures T (thermal gradient boundary condition) and mechanical pressures P as thermo-mechanical loads. Because there are not any changes including geometrical, loading and material dispersion in each circumferential angle, the 3D model of such cylinders can be reduced to a 2D axisymmetric model as shown in Fig. 1. It should be mentioned that axisymmetric models are able to deal with thick cylinders. However, shell theories which are only able to treat thin to moderately thick cylindrical shells need the consideration of too many assumptions that leads to losing the accuracy of the results. In additions, huge computational costs for 3D solutions of thick cylinders are needed. In the following modeling of TPGRs, a , b and their subscripts are utilized for inner and outer radii. In addition, L is the length of TPGRs.

2.1. Distributions of graphene nanosheets and pores

In the TPGRs, the distributions of graphene nanosheets along the radial direction are considered to be functionally graded. Given the considered axisymmetric model, graphene volume fraction does not change along the circumferential direction. In addition, graphene nanosheets with random orientations have been considered. To mathematically model such distributions, the following equation is used which shows how the volume fraction of graphene nanosheets $V_r(r)$ varies from V_{r-a} at the inner radius to V_{r-b} at the outer one [62]:

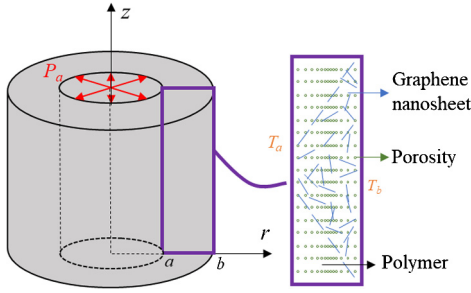


Fig. 1. FG distribution of graphene nanosheets and symmetrically distribution of porosity in axisymmetric TPGRs under thermal gradient effect and internal pressure.

$$V_r(r) = (V_{r-b} - V_{r-a}) \times [(r-a)/(b-a)]^p + V_{r-a} \quad (1)$$

where changing the scalar of p in the range of zero to infinity determines the pattern of graphene distribution along the thickness. Consequently, the volume fraction of polymeric matrix is $V_m = 1 - V_r$.

Furthermore, it is assumed that pores are symmetrically distributed along the radial direction of TPGRs. In the symmetric distribution patterns, the inner and outer surfaces of cylinders have no embedded pores, however, the highest concentration of pores can be seen at the mid thickness of cylinders. Using Gaussian random field technique for closed-cell cellular solids, the mass density ρ and Young's modulus E of TPGRs can be defined using a closed cell model based on those of graphene-reinforced nanocomposite (i.e. ρ^n and E^n) as [61,63]:

$$\rho = \left[1 - e_m \cos \left(\frac{\pi (r - (b+a)/2)}{(b-a)} \right) \right] \rho^n, \quad E = \left[1 - e_0 \cos \left(\frac{\pi (r - (b+a)/2)}{(b-a)} \right) \right] E^n \quad (2)$$

where e_0 is pore coefficient and $e_m = 1 - \sqrt{1 - e_0}$. In addition, Poisson's ratio of TPGRs ν is calculated as [55]:

$$\nu = 0.221\zeta + \nu^n (0.342\zeta^2 - 1.21\zeta + 1), \quad \zeta = 1 - \rho^p / \rho^n \quad (3)$$

To calculate the thermal expansion coefficient α of TPGRs an equation same as Eq. (2) has been utilized.

2.2. Material properties of nanocomposite

To compute the material properties of the nanocomposite considered in this paper, a revised form of Halpin-Tsai's approach capable of considering the geometrical shape of graphene nanosheets is employed. This revised HS's technique introduces two coefficients (η_1 and η_2) to the traditional micro-mechanical form of HS to compensate nanoscales interactions between graphene nanosheets and its polymeric matrix. Shen et al. [64] obtained these coefficients by matching the Yonge's modulus results obtained from traditional HS with those ones obtained from molecular dynamic (MD) modeling [6], both for graphene-reinforced nanocomposite. This revised HS's technique predicts Young's modulus of nanocomposite as follows [64]:

$$E^n = \frac{3}{8} \left[\eta_1 \frac{1 + 2(a^g/h^g)\gamma_1 V_r}{1 - \gamma_1 V_r} E^m \right] + \frac{5}{8} \left[\eta_2 \frac{1 + 2(b^g/h^g)\gamma_2 V_r}{1 - \gamma_2 V_r} E^m \right] \quad (4)$$

where

$$\gamma_1 = \frac{E_{11}^g/E^m - 1}{E_{11}^g/E^m + 2a^g/h^g}, \quad \gamma_2 = \frac{E_{22}^g/E^m - 1}{E_{22}^g/E^m + 2b^g/h^g} \quad (5)$$

where a^g , b^g and h^g are the dimensions of graphene. In addition, superscripts m and g denote the features of matrix and graphene, respectively. Moreover, the subscripts 11 and 22 are adopted to imply the longitudinal and transverse directions of graphene nanosheet, respectively.

Considering graphene-reinforced nanocomposite as a composite material reinforced by 3D randomly distributed short fibers, the thermal expansion coefficient $\alpha^n(r, T)$ of such nanocomposite can be calculated by employing the following equation as a function of radial location and temperature [65]:

$$\alpha^n(r, T) = \frac{[E^n + 4\nu^n(1 + \nu^n)K^n]\alpha_{11}^n + 4(1 + \nu^n)K^n\alpha_{22}^n}{E^n + 4(1 + \nu^n)^2 K^n} \quad (6)$$

where K^n is the bulk modulus of nanocomposite. In addition, the effective longitudinal and transverse of α^n are estimated as follows [64]:

$$\alpha_{11}^n(r, T) = \frac{V_r E_{11}^g \alpha_{11}^g + V_m E^m \alpha^m}{V_r E_{11}^g + V_m E^m} \quad (7)$$

$$\alpha_{22}^n(r, T) = (1 + \nu^g)V_r \alpha_{22}^g + (1 + \nu^m)V_m \alpha^m - \nu^n \alpha_{11}^n(r, T) \quad (8)$$

Furthermore, using the simple form of rule of mixture for the considered nanocomposite, effective density and Poisson's ratio can be approximated as:

$$\rho^n = V_r \rho^g + V_m \rho^m, \quad \nu^n = f_r \nu^g + f_m \nu^m \quad (9)$$

2.3. Basic equations

The constitutive law for such elastic materials makes a relationship between strain vector ϵ and stress vector σ using axisymmetric elasticity tensor $\mathbf{Q}$ based on Hook's law as follows:

$$\sigma = \mathbf{Q}\epsilon \quad (10)$$

where

$$\epsilon = [\epsilon_r \quad \epsilon_\theta \quad \epsilon_z \quad \epsilon_{rz}]^T, \quad \sigma = [\sigma_r \quad \sigma_\theta \quad \sigma_z \quad \sigma_{rz}]^T, \quad \mathbf{Q} = \frac{E}{(1 + \nu)(1 - 2\nu)} \begin{bmatrix} 1 - \nu & \nu & \nu & 0 \\ \nu & 1 - \nu & \nu & 0 \\ \nu & \nu & 1 - \nu & 0 \\ 0 & 0 & 0 & (1 - 2\nu)/2 \end{bmatrix} \quad (11)$$

in which θ shows the circumferential angle of TPGRs. Due to the consideration of concurrent thermal and mechanical loads, the deflection field of TPGRs is a summation of deflections caused by the mechanical load and the one caused by the thermal effect. Accordingly, the total strain vector ϵ of TPGRs can be obtained by summing up the thermal ϵ_T and mechanical ϵ portions of strain vectors as given below:

$$\epsilon = \epsilon + \epsilon_T \quad (12)$$

where the components of total and thermal strain vectors can be defined as follows:

$$e_r = \partial u_r / \partial r, \quad e_\theta = u_r / r, \quad e_z = \partial u_z / \partial z, \quad e_{rz} = (\partial u_r / \partial z) + (\partial u_z / \partial r) \quad (13)$$

$$\epsilon_T = \alpha \cdot \Delta T [1 \quad 1 \quad 1 \quad 0]^T \quad (14)$$

where u_z and u_r are the axial and radial deflections of TPGRs, respectively.

Moreover, the weak form of the equation of motion for such axisymmetric TPGRs is defined as below [44]:

$$\begin{aligned} \int_{\Omega} \sigma \cdot \delta(\epsilon) (2\pi r) d\Omega - \int_{\Gamma} \mathbf{F} \cdot \delta(\mathbf{u}) (2\pi R_i) d\Gamma \\ = - \int_{\Omega} \rho(z) \ddot{\mathbf{u}} \cdot \delta(\mathbf{u}) (2\pi r) d\Omega \end{aligned} \quad (15)$$

where $\mathbf{F}$, $\mathbf{u}$ and $\ddot{\mathbf{u}}$ are vectors of surface traction, displacement and acceleration, respectively. In addition, Γ is the axial length of TPGRs which mechanical load is enforced and Ω is the area of axisymmetric domain.

3. Meshless formulation

The main part of every meshless solutions is its interpolation function. In this paper, MLS interpolation function has been utilized to approximate the displacement field in TPGRs. The use of MLS interpolation function for such axisymmetric TPGRs with two degrees of freedom (u_r and u_z) for each node defines the deflection vector $\mathbf{u}$ as [44]:

$$\mathbf{u} = [u_r, u_z]^T = \Phi \hat{\mathbf{u}} \quad (16)$$

where the generalized values of deflection vector $\hat{\mathbf{u}}$ and the matrix of shape function Φ are described as:

$$\hat{\mathbf{u}} = [(\hat{u}_r)_1, (\hat{u}_z)_1, \dots, (\hat{u}_r)_N, (\hat{u}_z)_N]^T \quad (17)$$

$$\Phi = \begin{bmatrix} \Phi_1 & 0 & \Phi_2 & 0 & \dots & \dots & \Phi_N & 0 \\ 0 & \Phi_1 & 0 & \Phi_2 & \dots & \dots & 0 & \Phi_N \end{bmatrix} \quad (18)$$

where N is the number of nodes located inside the support domain. Substitution of Eq. (16) into Eq. (13) results in the definition of total strain vector in terms of generalized nodal values as:

$$\mathbf{e} = \mathbf{B} \hat{\mathbf{u}} \quad (19)$$

where

$$\mathbf{B} = \begin{bmatrix} \frac{\partial \Phi_1}{\partial r} & 0 & \frac{\partial \Phi_2}{\partial r} & 0 & \dots & \dots & \frac{\partial \Phi_N}{\partial r} & 0 \\ \frac{\partial \Phi_1}{\partial z} & 0 & \frac{\partial \Phi_2}{\partial z} & 0 & \dots & \dots & \frac{\partial \Phi_N}{\partial z} & 0 \\ 0 & \frac{\partial \Phi_1}{\partial r} & 0 & \frac{\partial \Phi_2}{\partial r} & \dots & \dots & 0 & \frac{\partial \Phi_N}{\partial r} \\ \frac{\partial \Phi_1}{\partial z} & \frac{\partial \Phi_1}{\partial r} & \frac{\partial \Phi_2}{\partial z} & \frac{\partial \Phi_2}{\partial r} & \dots & \dots & \frac{\partial \Phi_N}{\partial z} & \frac{\partial \Phi_N}{\partial r} \end{bmatrix} \quad (20)$$

Finally, introducing the definitions of stress (Eq. (10)), strain (Eq. (19)) and deflection (Eq. (16)) vectors to in the weak form of governing equation (Eq. (15)) gives the following standard form of the system of equations:

$$\mathbf{M} \ddot{\mathbf{u}} + \mathbf{C} \dot{\mathbf{u}} + \mathbf{k} \mathbf{u} = \mathbf{f} + \mathbf{f}_T \quad (21)$$

where $\mathbf{f}$ and $\mathbf{f}_T$ are surface traction vector and thermal load vector, respectively. Moreover, $\mathbf{M}$, $\mathbf{C}$, $\mathbf{k}$ are the system matrices of mass, damping and stiffness, respectively. Considering structural (Rayleigh) damping for the system, these vectors and matrices defined in Eq. (19) are as follows:

$$\begin{aligned} \mathbf{f} &= \int_{\Gamma} \Phi^T \mathbf{F} ds, \quad \mathbf{f}_T = \int_{\Omega} \mathbf{B}^T \mathbf{Q} \epsilon_T dv \mathbf{M} = \int_{\Omega} \rho \Phi^T \Phi dv, \\ \mathbf{C} &= C_m \mathbf{M} + C_k \mathbf{k}, \quad \mathbf{k} = \int_{\Omega} \mathbf{B}^T \mathbf{Q} \mathbf{B} dv \end{aligned} \quad (22)$$

where inertial and stiffness coefficients of Rayleigh structural damping are shown with C_m and C_k , respectively.

4. Results and discussions

The multifunctional lightweight cylinders considered in this paper are supposed to be made of a polymeric nanocomposite material with symmetrically distributed embedded pores inside it where the nanocomposite is a combination of well-dispersed PMMA and randomly oriented graphene nanosheets (with $h^g = 0.188$ nm, $a^g = 14.76$ nm and $b^g = 14.77$ nm [64]). For the considered PMMA and graphene nanosheets, mass density and Poisson's ratio are considered as follows [64]: $\rho^g = 4118$ kg/m³, $\rho^m = 1150$ kg/m³, $\nu^g = 0.177$ and $\nu^m = 0.34$. However, for the both components of the considered nanocomposite, thermal conductivity and Young's modulus are assumed as functions of temperature which can be determined as follows [64]:

$$\begin{aligned} \alpha^m &= 45(1 + 0.0005\Delta T) \times 10^{-6}/\text{K}, \\ \alpha_{11}^g &= (-3.83788 - 0.01416T - 1.63355 \times 10^{-5}T^2 \\ &\quad + 6.33589 \times 10^{-9}T^3) \times 10^{-6}/\text{K} \\ \alpha_{22}^g &= (-3.73997 - 0.01296T - 1.35033 \times 10^{-5}T^2 \\ &\quad + 4.60392 \times 10^{-9}T^3) \times 10^{-6}/\text{K} \\ E^m &= (3.52 - 0.0034T) \text{ GPa}, \\ E_{11}^g &= (2.16637 - 0.00193T + 2.93701 \times 10^{-6}T^2 \\ &\quad - 1.51775 \times 10^{-9}T^3) \text{ TPa} \\ E_{22}^g &= (2.16868 - 0.00193T + 2.85954 \times 10^{-6}T^2 \\ &\quad - 1.45145 \times 10^{-9}T^3) \text{ TPa} \end{aligned} \quad (23)$$

Furthermore, the values of η_1 and η_2 defined in Eq. (4) are presented in Table 1 [64] for different volume fractions of nanofiller at some temperatures. For the other values of f_r and T , the values of η_1 and η_2 are interpolated using a 2D spline function based on the values of Table 1.

4.1. Validation study

Consider long axisymmetric cylinders made of an isotropic and homogeneous material subjected to a steady state thermal gradient effect, or static internal P_a and external P_b pressures. Hetnarski and Eslami [66] presented exact solutions for the radial and circumferential stress distributions of such cylinders as follows:

Under thermal gradient effect:

$$\begin{cases} \sigma_r = S_t \left[\ln \frac{b}{r} + \frac{a^2}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right) \ln \frac{b}{a} \right] \\ \sigma_\theta = S_t \left[1 - \ln \frac{b}{r} - \frac{a^2}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right) \ln \frac{b}{a} \right] \end{cases} \quad (25)$$

Under mechanical load:

$$\begin{cases} \sigma_r = S_{ma} \left(1 - \frac{b^2}{r^2} \right) - S_{mb} \left(1 - \frac{a^2}{r^2} \right) \\ \sigma_\theta = S_{ma} \left(1 + \frac{b^2}{r^2} \right) - S_{mb} \left(1 + \frac{a^2}{r^2} \right) \end{cases} \quad (26)$$

Table 1
 η_1 and η_2 utilized in HS's equation for the graphene-reinforced nanocomposite [64].

T (K)	η	f_r (%)				
		3	5	7	9	11
300	η_1	2.929	3.068	3.013	2.647	2.311
	η_2	2.855	2.962	2.966	2.609	2.260
400	η_1	2.977	3.128	3.060	2.701	2.405
	η_2	2.896	3.023	3.027	2.603	2.337
500	η_1	3.388	3.544	3.462	3.058	2.736
	η_2	3.382	3.414	3.339	2.936	2.665

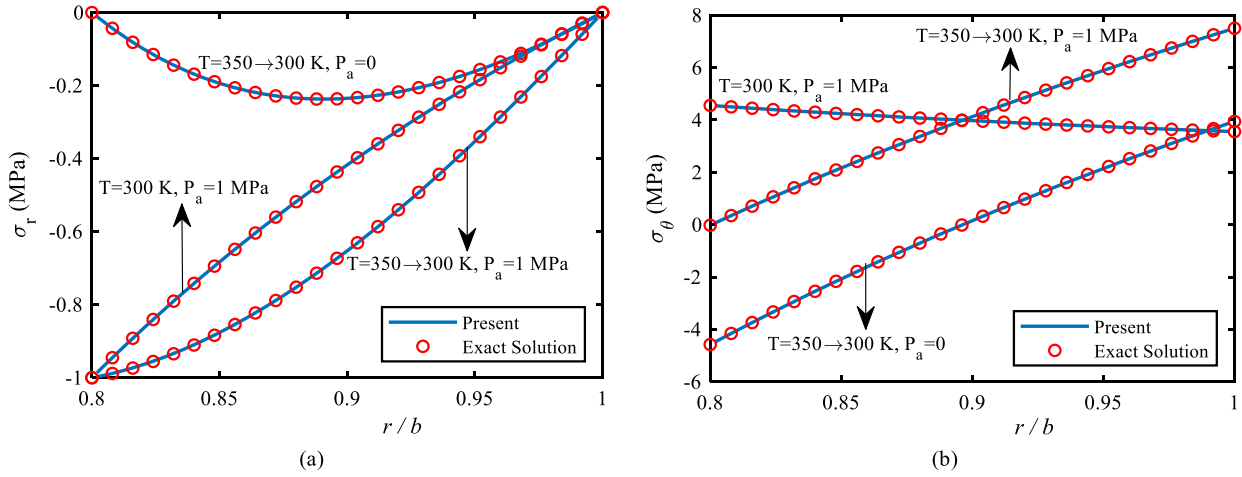


Fig. 2. (a) σ_r and (b) σ_θ responses of a long axisymmetric cylinder made of PMMA under thermo-mechanical loads, a comparison study.

where

$$S_t = \frac{E\alpha(T_a - T_b)}{2(1-\nu)\ln\frac{b}{a}}, \quad S_{ma} = \frac{P_a a^2}{b^2 - a^2},$$

$$S_{mb} = \frac{P_b b^2}{b^2 - a^2} \quad (27)$$

In view of the novelty of the current work, the reliability of our developed meshless method can be examined by making a comparison study between our solution and the exact ones (Eq. X). With this regard, a long axisymmetric cylinder made of PMMA with $E = 2.5$ GPa, $\nu = 0.34$ and $\alpha = 45 \times 10^{-6}/K$ under thermal gradient effect ($T_a = 350$ K and $T_b = 300$ K), mechanical load ($T = 300$ K, $P_a = 1$ MPa) or thermo-mechanical load ($T_a = 350$ K, $T_b = 300$ K and $P_a = 1$ MPa) is considered. Fig. 2 shows the comparisons between our solution and the exact ones on the radial and circumferential stress distributions of such cylinder. A perfect consistency between the results is observed which demonstrates the reliability of the meshless results in this work.

4.2. Temperature response in TPGRG

Since thermal gradient environment have been considered in this paper, the steady state temperature response of such cylinders is investigated in this subsection. The effect of two main parameters including pore coefficient, and the distribution of graphene nanosheets on the temperature profile through the thickness of such cylinders are studied in Fig. 3. As shown in Fig. 3a, when graphene nanosheets are dispersed uniformly, the increase of pore coefficient leads to lower thermal conductivity around the middle thickness of cylinders due to the symmetric distribution of pores. As a result, temperature profiles in porous cylinders have higher nonlinearity forms. Moreover, it can be seen that the distribution

of graphene nanosheets has a slight impact on the temperature profile of TPGRGs.

4.3. Stress wave in TPGRG

The propagations of stress waves and deflections through the thickness of the proposed thick porous graphene-reinforced cylinders are investigated in this subsection. The results are presented in as functions of time to show the differences between the waves of deflections (vibrations) and stresses. In this regard, a clamped-clamped axisymmetric TPGRG with the default specifications of $b = 0.2$ m, $a/b = 0.75$, $L/b = 2$, $T_a = 350$ K, $T_b = 300$ K, $V_{r-a} = 3\%$, $V_{r-b} = 11\%$, $p = 1$, $e_0 = 0.6$, $C_m = 3000$ and $C_k = 3 \times 10^{-6}$ is considered for the following modelings, unless otherwise clearly mentioned. To study the dynamic behavior of such cylinder, the deflection response of the same cylinders under the static load of $P_a = 20$ MPa is considered as the initial condition of dynamic analyses.

In the first modeling, the effect of damping coefficients on the propagations of stress waves and deflections is investigated in Fig. 4. As expected, results show that the increase of damping coefficients leads to significant reductions in the converging (stationary) times. The dynamic results also show that the alternation of damping coefficient does not affect the propagation speeds and the initial values of dynamic responses. As shown, the dynamic responses are compared with the static responses of the same cylinder under the same steady state thermal gradient load (i.e. $T_a = 350$ K, $T_b = 300$ K and $P_a = 0$). The comparison of static and dynamic results discloses that the dynamic responses are exactly converged to the responses of static solution. This consistency between static and dynamic results can be also taken as another verification of the present results.

The impact of embedding void inside the nanocomposite cylinders on the dynamic responses of TPGRGs is presented in Fig. 5.

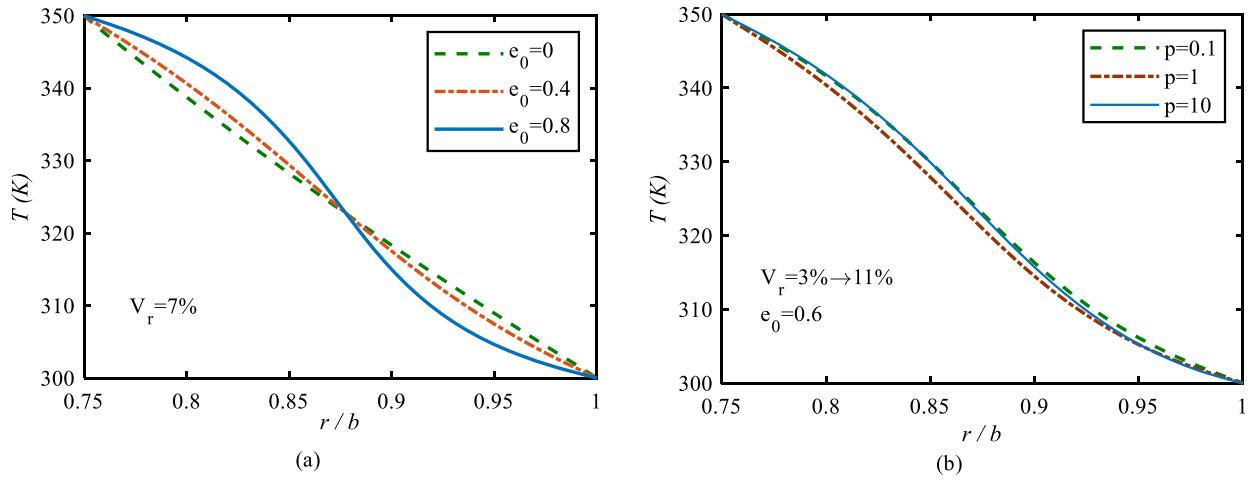


Fig. 3. Temperature profile through the thickness of TPGRC for different (a) pore coefficient (b) distributions of graphene nanosheets.

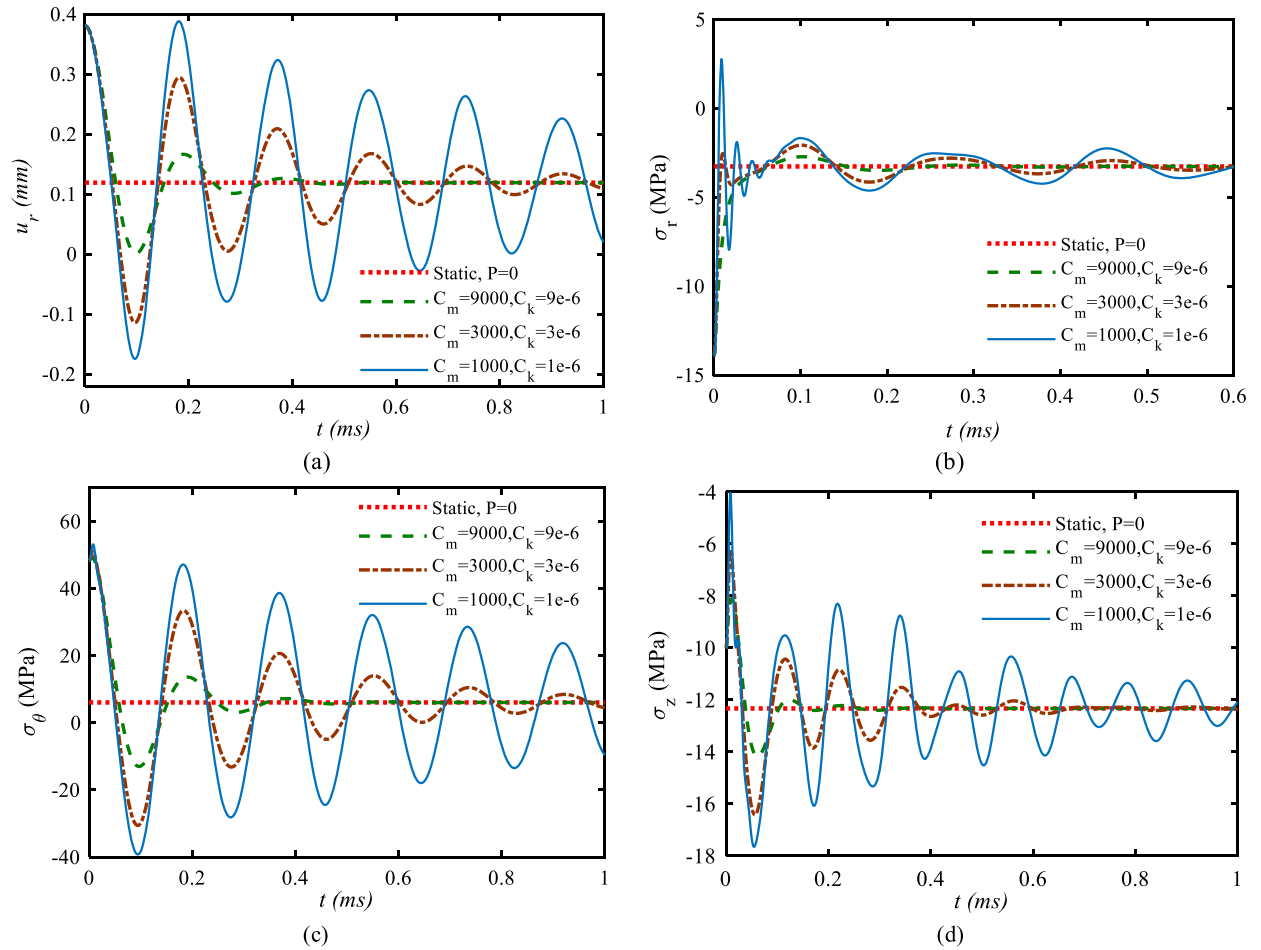


Fig. 4. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ and (d) σ_z at the mid-point ($r = (a+b)/2$ and $z = L/2$) of an axisymmetric TPGRC with different damping coefficients in comparison with static response.

Fig. 5a illustrates that the increase of porosity parameter significantly increases the amplitudes and stationary time of vibrations; however, it reduces the values of stationary deflections because embedding pores leads to the improvement of structural stiffness, but the reduction of structural thermal conductivity in TPGRCs. The propagations of stress waves also show the increases of stress values along with the increase of propagation speed as e_0 reduces. Due to having clamped supports at the ends of TPGRCs, the values of axial stresses are significant, especially for the perfect nanocom-

posite cylinder ($e_0 = 0$). However, the values of shear stress are very low and e_0 does not have a considerable effect on its propagation. In addition, the reductions of the stationary values of radial and axial stresses are observed as e_0 increases.

In Fig. 6, the effect of thermal load on the propagations of stresses and radial deflections at the mid-point of the considered TPGRC is illustrated by changing temperatures at inner and outer surfaces. When there is no thermal effect ($T = 300$ K), Fig. 6 shows that the dynamic responses of radial deflections and stresses are

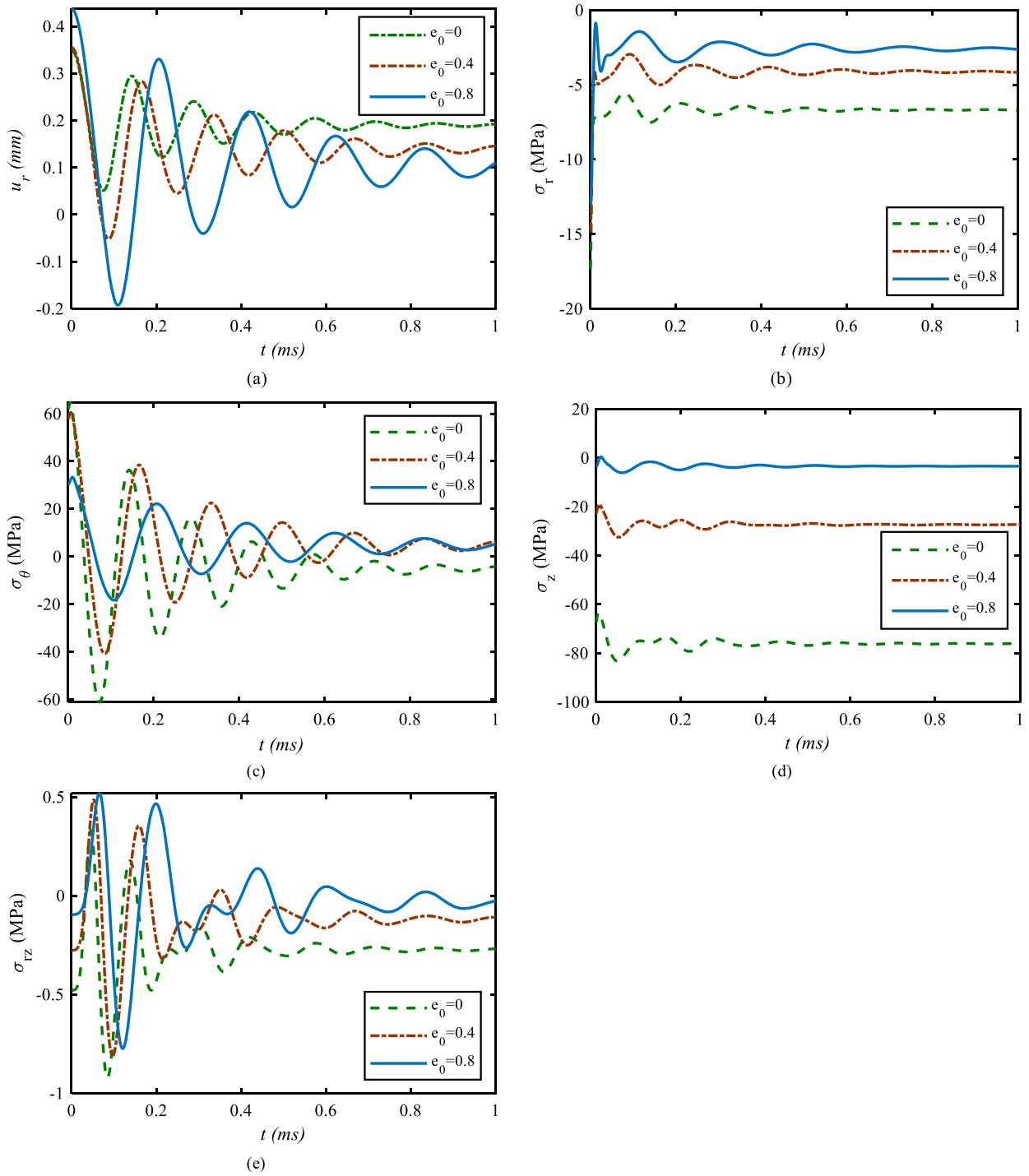


Fig. 5. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ , (d) σ_z and (e) σ_{rz} at the mid-point of axisymmetric TPGRCs with $e_0 = 0, 0.4, 0.8$.

oscillated around zero and converged to that as the effect of initial condition is being damped by the considered structural damping. However, thermal load significantly changes the stationary values of deflections and stresses such that cylinders with higher temperatures have higher those stationary values. In addition, rising temperature in the cylinder slightly reduces propagation speeds, but it slightly amplifies the amplitudes of vibrations and stress waves because rising the temperature reduces the Young's modulus of TPGRC. Furthermore, Fig. 6 shows that the responses of cylinders under thermal gradient effects of $T_a = 300$ K and $T_b = 400$ K and

vice versa are always between the dynamic responses of cylinders with constant temperatures of $T = 300$ K and $T = 400$ K.

Figs. 7 and 8 show the influences of graphene volume fraction and distribution on the propagations of deflection and stress waves of axisymmetric TPGRCs, respectively. Fig. 7a discloses that there is a remarkable difference between the time history of radial deflections in a polymeric cylinder ($V_r = 0$) and cylinders reinforced with uniformly distributed graphene nanosheets such that the deflections and stationary time of the polymeric cylinder are much higher than those of nanocomposite cylinders. However, the propagation speeds of stress waves and deflections in polymeric

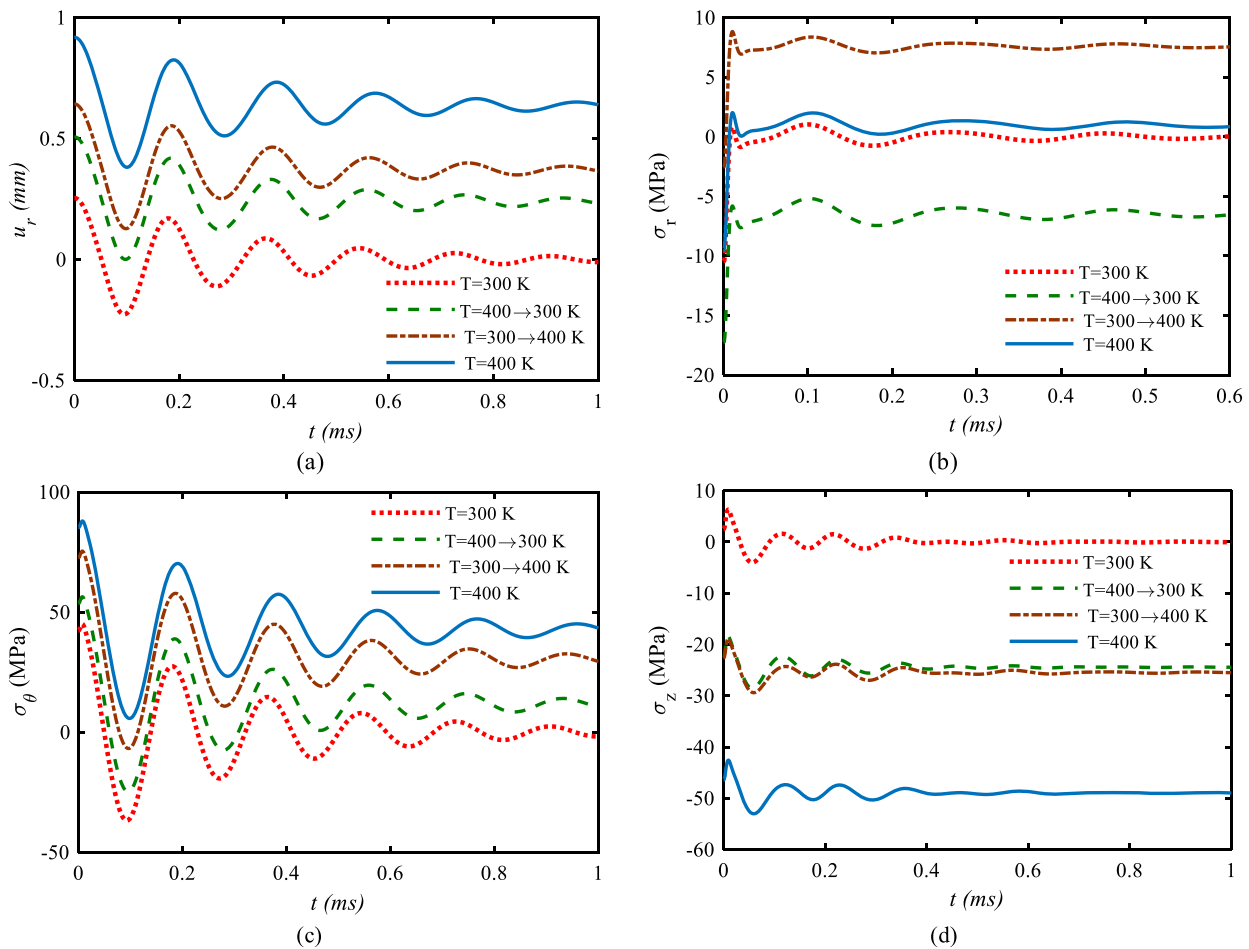


Fig. 6. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ and (d) σ_z at the mid-point of an axisymmetric TPGRc under different thermal loads.

cylinder are much less than these speeds in nanocomposite cylinders. In addition, the use of higher amount of graphene nanosheets in nanocomposite cylinders leads to the increase of the values and propagation speeds of stress waves as plotted in Fig. 7.

The considerable effect of nanofiller distribution on the propagations of deflections and stress waves are shown in Fig. 8. Fig. 8a discloses that the amplitude of deflections is intensified by the increase of p , while the propagation speed is reduced. In the propagations of stress waves, Figs. 8b–8d show the values of stresses and their propagation speeds are increased as p is reduced. It should be mentioned that the increase of p not only changes the profile of the distribution of graphene nanosheets but also it decreases the amount of such nanofillers inside the cylinders. The comparison between the dynamic response of TPGRc with $p = 1$ and TPGRc with UD of nanofillers ($V_r = 7\%$) shows that the propagation speed and values of stresses in UD cylinder are higher than those of the FG one.

Finally, the effect of thickness on the propagations of deflections and stress waves is investigated by changing inner radius (a) of TPGRcs in Fig. 9. As shown in Fig. 9a, thinner TPGRcs have much higher initial deflection, but their stationary deflection is a little higher than that of thicker TPGRcs. In addition, the propagation speeds of deflections and stress waves in thicker TPGRcs are higher than those of thinner TPGRcs. Figs. 9b–9d also show that the values of stresses and their amplitudes in thinner cylinders are higher than those of thicker ones.

5. Conclusions

The propagations of stress waves and deflections in thick porous graphene-reinforced cylinders under thermal gradient effects were investigated using a developed meshless solution. FG profiles and a symmetric pattern were considered for the distributions of graphene nanofillers and pores along their radial direction, respectively. Considering structural damping for the system, the influences of major involved parameters of TPGRcs such as effects of porosity, thermal load, and graphene on the stress wave responses of TPGRcs were studied. It was found that embedding more voids significantly increases the amplitudes and stationary time of vibrations; however, it reduces the values of stationary deflections and stresses, along with the reduction of propagation speeds. Moreover, it was seen that reinforcing pure polymeric cylinders with graphene nanosheets causes remarkable reductions in the deflections and stationary time of the resulted nanocomposite cylinders. Furthermore, propagation speeds in pure polymeric cylinder is much less than those of nanocomposite cylinders. Finally, the results showed that rising the temperature of TPGRcs slightly reduces propagation speeds.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

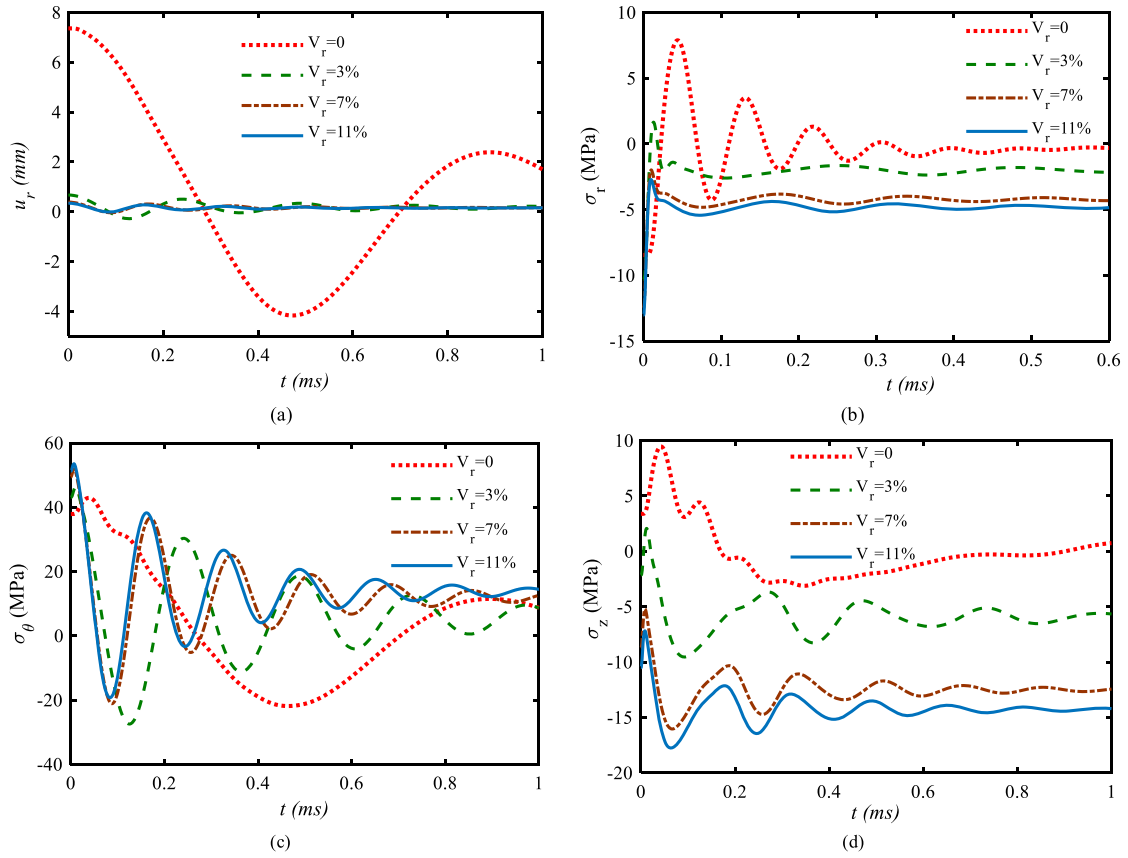


Fig. 7. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ and (d) σ_z at the mid-point of axisymmetric TPGRs with different amounts of graphene nanosheets.

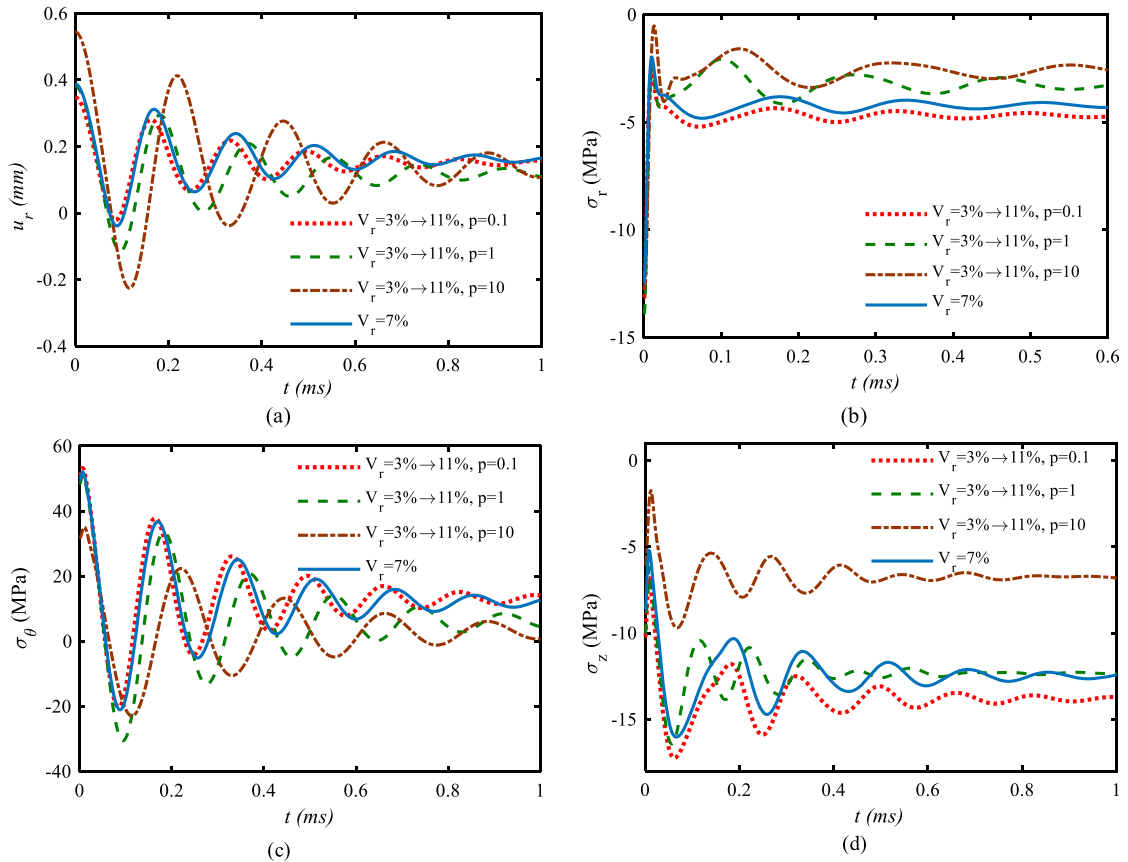


Fig. 8. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ and (d) σ_z at the mid-point of axisymmetric TPGRs with different distributions of graphene nanosheets.

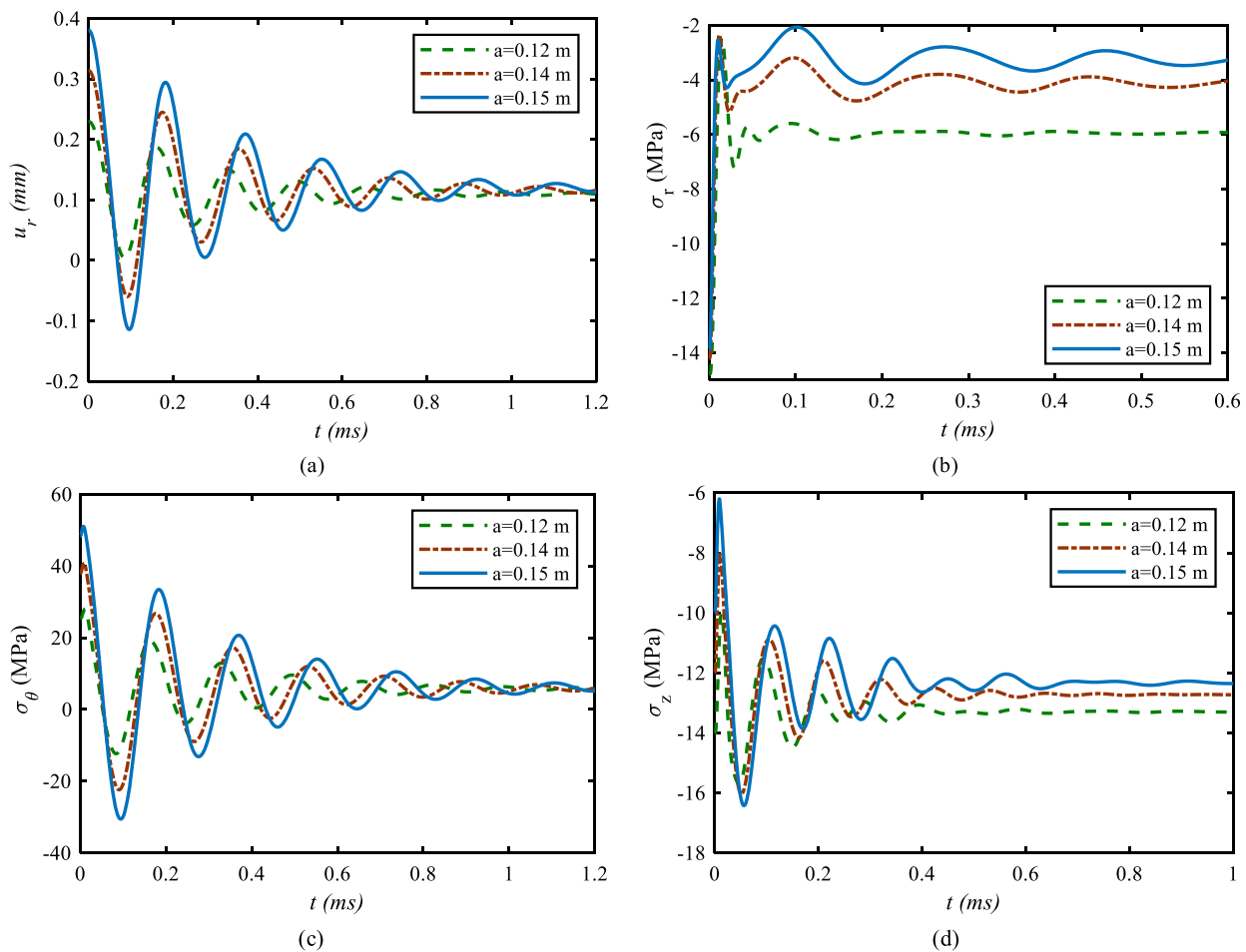


Fig. 9. Time history responses of (a) U_r , (b) σ_r , (c) σ_θ and (d) σ_z at the mid-point of axisymmetric TPGRCS with different thicknesses.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

Acknowledgements

The work described in this paper was supported by Natural Sciences and Engineering Research Council of Canada (NSERC under grant RGPIN-217525). The authors are grateful for their supports.

References

- [1] M.F. Caliri, A.J.M. Ferreira, V. Tita, A review on plate and shell theories for laminated and sandwich structures highlighting the finite element method, *Compos. Struct.* 156 (2016) 63–77, <https://doi.org/10.1016/j.compstruct.2016.02.036>.
- [2] P. Ma, N.A. Siddiqui, G. Marom, J. Kim, Dispersion and functionalization of carbon nanotubes for polymer-based nanocomposites: a review, *Composites, Part A, Appl. Sci. Manuf.* 41 (2010) 1345–1367, <https://doi.org/10.1016/j.compositesa.2010.07.003>.
- [3] Y. Zare, K.Y. Rhee, Simulation of tensile modulus of polymer carbon nanotubes nanocomposites in the case of incomplete interfacial bonding between polymer matrix and carbon nanotubes by critical interfacial parameters, *Polym. Commun. (Guildf.)* 191 (2020) 122260, <https://doi.org/10.1016/j.polymer.2020.122260>.
- [4] S. Karimzadeh, B. Safaei, T.C. Jen, Investigate the importance of mechanical properties of SWCNT on doxorubicin anti-cancer drug adsorption for medical application: a molecular dynamic study, *J. Mol. Graph. Model.* 101 (2020) 1–29, <https://doi.org/10.1016/j.jmgm.2020.107745>.
- [5] A.A. Balandin, Thermal properties of graphene and nanostructured carbon materials, *Nat. Mater.* 10 (2011) 569–581, <https://doi.org/10.1038/nmat3064>.
- [6] F. Lin, Y. Xiang, H.S. Shen, Temperature dependent mechanical properties of graphene reinforced polymer nanocomposites – a molecular dynamics simulation, *Composites, Part B, Eng.* 111 (2017) 261–269, <https://doi.org/10.1016/j.compositesb.2016.12.004>.
- [7] M. Foroutan, R. Moradi-Dastjerdi, R. Sotoodeh-Bahreini, Static analysis of FGM cylinders by a mesh-free method, *Steel Compos. Struct.* 12 (2012) 1–11, <https://doi.org/10.12989/scs.2011.12.1.001>.
- [8] M.-C. Trinh, D.-D. Nguyen, S.-E. Kim, T.M. Chien, N.D. Duc, S.K. Eock, Effects of porosity and thermomechanical loading on free vibration and nonlinear dynamic response of functionally graded sandwich shells with double curvature, *Aerosp. Sci. Technol.* 87 (2019) 119–132, <https://doi.org/10.1016/j.ast.2019.02.010>.
- [9] A. Khandan, E. Nassireslami, S. Saber-Samandari, N. Arabi, Fabrication and characterization of porous bioceramic-magnetite biocomposite for maxillofacial fractures application, *Dent. Hypoth.* 11 (2020) 78–85, <https://doi.org/10.4103/denthyp.denthyp>.
- [10] B. Safaei, The effect of embedding a porous core on the free vibration behavior of laminated composite plates, *Steel Compos. Struct.* 35 (2020) 659–670, <https://doi.org/10.12989/scs.2020.35.5.659>.
- [11] R. Moradi-Dastjerdi, K. Behdinan, B. Safaei, Z. Qin, Static performance of agglomerated CNT-reinforced porous plates bonded with piezoceramic faces, *Int. J. Mech. Sci.* 188 (2020) 105966, <https://doi.org/10.1016/j.ijmecsci.2020.105966>.
- [12] R. Moradi-Dastjerdi, K. Behdinan, B. Safaei, Z. Qin, Buckling behavior of porous CNT-reinforced plates integrated between active piezoelectric layers, *Eng. Struct.* 222 (2020) 111141, <https://doi.org/10.1016/j.engstruct.2020.111141>.
- [13] M. Esmailzadeh, M. Kadkhodayan, Dynamic analysis of stiffened bi-directional functionally graded plates with porosities under a moving load by dynamic relaxation method with kinetic damping, *Aerosp. Sci. Technol.* 93 (2019) 105333, <https://doi.org/10.1016/j.ast.2019.105333>.
- [14] A.R. Setoodeh, M. Shojaee, P. Malekzadeh, Vibrational behavior of doubly curved smart sandwich shells with FG-CNTRC face sheets and FG porous core, *Composites, Part B, Eng.* 165 (2019) 798–822, <https://doi.org/10.1016/j.compositesb.2019.01.022>.
- [15] L.B. Nguyen, C.H. Thai, A.M. Zenkour, H. Nguyen-Xuan, An isogeometric Bézier finite element method for vibration analysis of functionally graded piezoelec-

- tric material porous plates, *Int. J. Mech. Sci.* 157–158 (2019) 165–183, <https://doi.org/10.1016/j.ijmecsci.2019.04.017>.
- [16] R. Moradi-Dastjerdi, K. Behdinan, Thermo-electro-mechanical behavior of an advanced smart lightweight sandwich plate, *Aerosp. Sci. Technol.* 106 (2020) 106142, <https://doi.org/10.1016/j.ast.2020.106142>.
 - [17] W. Zhang, Y. Niu, K. Behdinan, Vibration characteristics of rotating pretwisted composite tapered blade with graphene coating layers, *Aerosp. Sci. Technol.* 98 (2020) 105644, <https://doi.org/10.1016/j.ast.2019.105644>.
 - [18] Y. Sun, G. Shi, Graphene/polymer composites for energy applications, *J. Polym. Sci., Part B, Polym. Phys.* 51 (2013) 231–253, <https://doi.org/10.1002/polb.23226>.
 - [19] Y.H. Yang, L. Bolling, M.A. Priolo, J.C. Grunlan, Super gas barrier and selectivity of graphene oxide-polymer multilayer thin films, *Adv. Mater.* 25 (2013) 503–508, <https://doi.org/10.1002/adma.201202951>.
 - [20] Y. Cui, S.I. Kundalwal, S. Kumar, Gas barrier performance of graphene/polymer nanocomposites, *Carbon N.Y.* 98 (2016) 313–333, <https://doi.org/10.1016/j.carbon.2015.11.018>.
 - [21] M.A.L. Machado, L. Valentini, J. Biagiotti, J.M. Kenny, Thermal and mechanical properties of single-walled carbon nanotubes-polypropylene composites prepared by melt processing, *Carbon N.Y.* 43 (2005) 1499–1505, <https://doi.org/10.1016/j.carbon.2005.01.031>.
 - [22] A.R. Alian, S.A. Meguid, S.I. Kundalwal, Unraveling the influence of grain boundaries on the mechanical properties of polycrystalline carbon nanotubes, *Carbon N.Y.* 125 (2017) 180–188, <https://doi.org/10.1016/j.carbon.2017.09.056>.
 - [23] M.A. Rafiee, J. Rafiee, Z.Z. Yu, N. Koratkar, Buckling resistant graphene nanocomposites, *Appl. Phys. Lett.* 95 (2009) 223103, <https://doi.org/10.1063/1.3269637>.
 - [24] A.R. Alian, S.A. Meguid, Hybrid molecular dynamics-finite element simulations of the elastic behavior of polycrystalline graphene, *Int. J. Mech. Mater. Des.* 14 (2018) 551–563, <https://doi.org/10.1007/s10999-017-9389-y>.
 - [25] A.M. Fattahi, B. Safaei, N.A. Ahmed, A comparison for the non-classical plate model based on axial buckling of single-layered graphene sheets, *Eur. Phys. J. Plus* 134 (2019) 555, <https://doi.org/10.1140/epjp/i2019-12912-7>.
 - [26] S. Zhao, Z. Zhao, Z. Yang, L.L. Ke, S. Kitipornchai, J. Yang, Functionally graded graphene reinforced composite structures: a review, *Eng. Struct.* 210 (2020) 110339, <https://doi.org/10.1016/j.engstruct.2020.110339>.
 - [27] N.P. Singh, V.K. Gupta, A.P. Singh, Graphene and carbon nanotube reinforced epoxy nanocomposites: a review, *Polym. Commun. (Guildf.)* 180 (2019) 121724, <https://doi.org/10.1016/j.POLYMER.2019.121724>.
 - [28] K. Mehar, S.K. Panda, Nonlinear finite element solutions of thermoelastic flexural strength and stress values of temperature dependent graded CNT-reinforced sandwich shallow shell structure, *Struct. Eng. Mech.* 67 (2018) 565–578, <https://doi.org/10.12989/sem.2018.67.6.565>.
 - [29] K. Behdinan, R. Moradi-Dastjerdi, B. Safaei, Z. Qin, F. Chu, D. Hui, Graphene and CNT impact on heat transfer response of nanocomposite cylinders, *Nanotechnol. Rev.* 9 (2020) 41–52, <https://doi.org/10.1515/ntrev-2020-0004>.
 - [30] A. Pourasghar, Z. Chen, Hyperbolic heat conduction and thermoelastic solution of functionally graded CNT reinforced cylindrical panel subjected to heat pulse, *Int. J. Solids Struct.* 163 (2019) 117–129, <https://doi.org/10.1016/j.ijsolstr.2018.12.030>.
 - [31] A. Alibeigloo, Thermoelastic analysis of functionally graded carbon nanotube reinforced composite cylindrical panel embedded in piezoelectric sensor and actuator layers, *Composites, Part B, Eng.* 98 (2016) 225–243, <https://doi.org/10.1016/j.compositesb.2016.05.010>.
 - [32] H.S. Shen, Y. Xiang, Y. Fan, D. Hui, Nonlinear bending analysis of FG-GRC laminated cylindrical panels on elastic foundations in thermal environments, *Composites, Part B, Eng.* 141 (2018) 148–157, <https://doi.org/10.1016/j.compositesb.2017.12.048>.
 - [33] B. Yang, S. Kitipornchai, Y.-F. Yang, J. Yang, 3D thermo-mechanical bending solution of functionally graded graphene reinforced circular and annular plates, *Appl. Math. Model.* 49 (2017) 69–86, <https://doi.org/10.1016/j.apm.2017.04.044>.
 - [34] H. Asadi, Numerical simulation of the fluid-solid interaction for CNT reinforced functionally graded cylindrical shells in thermal environments, *Acta Astronaut.* 138 (2017) 214–224, <https://doi.org/10.1016/j.actaastro.2017.05.039>.
 - [35] B. Sobhaniragh, R.C. Batra, W.J. Mansur, F.C. Peters, Thermal response of ceramic matrix nanocomposite cylindrical shells using Eshelby-Mori-Tanaka homogenization scheme, *Composites, Part B, Eng.* 118 (2017) 41–53, <https://doi.org/10.1016/j.compositesb.2017.02.032>.
 - [36] A. Pourasghar, Z. Chen, Thermoelastic response of CNT reinforced cylindrical panel resting on elastic foundation using theory of elasticity, *Composites, Part B, Eng.* 99 (2016) 436–444, <https://doi.org/10.1016/j.compositesb.2016.06.028>.
 - [37] P. Malekzadeh, A.R. Setoodeh, M. Shojaei, Vibration of FG-GPLs eccentric annular plates embedded in piezoelectric layers using a transformed differential quadrature method, *Comput. Methods Appl. Mech. Eng.* 340 (2018) 451–479, <https://doi.org/10.1016/j.cma.2018.06.006>.
 - [38] X. Liu, K. Behdinan, A novel analytical model for the thermal behavior of a fiber-reinforced plastic pulley in a front-end accessory drive, *Adv. Mech. Eng.* 12 (2020), <https://doi.org/10.1177/1687814020924492>.
 - [39] R. Gholami, R. Ansari, Nonlinear harmonically excited vibration of third-order shear deformable functionally graded graphene platelet-reinforced composite rectangular plates, *Eng. Struct.* 156 (2018) 197–209, <https://doi.org/10.1016/j.engstruct.2017.11.019>.
 - [40] R. Gholami, R. Ansari, On the nonlinear vibrations of polymer nanocomposite rectangular plates reinforced by graphene nanoplatelets: a unified higher-order shear deformable model, *Iran. J. Sci. Technol. Trans. Mech. Eng.* 43 (2019) 603–620, <https://doi.org/10.1007/s40997-018-0182-9>.
 - [41] R. Moradi-Dastjerdi, K. Behdinan, Thermoelastic static and vibrational behaviors of nanocomposite thick cylinders reinforced with graphene, *Steel Compos. Struct.* 31 (2019) 529–539, <https://doi.org/10.12989/scs.2019.31.5.529>.
 - [42] R. Moradi-Dastjerdi, G. Payganeh, M. Tajdari, Thermoelastic analysis of functionally graded cylinders reinforced by wavy CNT using a mesh-free method, *Polym. Compos.* 39 (2018) 2190–2201, <https://doi.org/10.1002/pc.24183>.
 - [43] M. Song, S. Kitipornchai, J. Yang, Free and forced vibrations of functionally graded polymer composite plates reinforced with graphene nanoplatelets, *Compos. Struct.* 159 (2017) 579–588, <https://doi.org/10.1016/j.compstruct.2016.09.070>.
 - [44] R. Moradi-Dastjerdi, G. Payganeh, M. Tajdari, Resonance in functionally graded nanocomposite cylinders reinforced by wavy carbon nanotube, *Polym. Compos.* 38 (2017) E542–E552, <https://doi.org/10.1002/pc.24045>.
 - [45] R. Moradi-Dastjerdi, G. Payganeh, Thermoelastic dynamic analysis of wavy carbon nanotube reinforced cylinders under thermal loads, *Steel Compos. Struct.* 25 (2017) 315–326, <https://doi.org/10.12989/scs.2017.25.3.315>.
 - [46] Y. Fan, Y. Xiang, H.-S. Shen, Nonlinear forced vibration of FG-GRC laminated plates resting on visco-Pasternak foundations, *Compos. Struct.* 209 (2019) 443–452, <https://doi.org/10.1016/j.compstruct.2018.10.084>.
 - [47] H.P. Cong, N.D. Duc, New approach to investigate the nonlinear dynamic response and vibration of a functionally graded multilayer graphene nanocomposite plate on a viscoelastic Pasternak medium in a thermal environment, *Acta Mech.* 229 (2018) 3651–3670, <https://doi.org/10.1007/s00707-018-2178-3>.
 - [48] S.M. Hosseini, C. Zhang, Elastodynamic and wave propagation analysis in a FG graphene platelets-reinforced nanocomposite cylinder using a modified nonlinear micromechanical model, *Steel Compos. Struct.* 27 (2018) 255–271.
 - [49] Z. Qin, B. Safaei, X. Pang, F. Chu, Traveling wave analysis of rotating functionally graded graphene platelet reinforced nanocomposite cylindrical shells with general boundary conditions, *Results Phys.* 15 (2019) 102752, <https://doi.org/10.1016/j.rinp.2019.102752>.
 - [50] M. Babaei, M. Hadi, K. Asemi, Natural frequency and dynamic analyses of functionally graded saturated porous annular sector plate and cylindrical panel based on 3D elasticity, *Aerosp. Sci. Technol.* 96 (2020) 105524, <https://doi.org/10.1016/j.ast.2019.105524>.
 - [51] R. Moradi-Dastjerdi, K. Behdinan, Temperature effect on free vibration response of a smart multifunctional sandwich plate, *J. Sandw. Struct. Mater.* (2020), <https://doi.org/10.1177/1099636220908707>.
 - [52] R. Moradi-Dastjerdi, K. Behdinan, Stability analysis of multifunctional smart sandwich plates with graphene nanocomposite and porous layers, *Int. J. Mech. Sci.* 167 (2020) 105283, <https://doi.org/10.1016/j.ijmecsci.2019.105283>.
 - [53] R. Moradi-Dastjerdi, A. Radhi, K. Behdinan, Damped dynamic behavior of an advanced piezoelectric sandwich plate, *Compos. Struct.* 243 (2020) 112243, <https://doi.org/10.1016/j.compstruct.2020.112243>.
 - [54] B. Anirudh, M. Ganapathi, C. Anant, O. Polit, A comprehensive analysis of porous graphene-reinforced curved beams by finite element approach using higher-order structural theory: bending, vibration and buckling, *Compos. Struct.* 222 (2019) 110899, <https://doi.org/10.1016/j.compstruct.2019.110899>.
 - [55] L.B. Nguyen, N.V. Nguyen, C.H. Thai, A.M.J. Ferreira, H. Nguyen-xuan, An isogeometric Bézier finite element analysis for piezoelectric FG porous plates reinforced by graphene platelets, *Compos. Struct.* 214 (2019) 227–245, <https://doi.org/10.1016/j.compstruct.2019.01.077>.
 - [56] M.R. Barati, A.M. Zenkour, Vibration analysis of functionally graded graphene platelet reinforced cylindrical shells with different porosity distributions, *Mech. Adv. Mat. Struct.* 26 (2019) 1580–1588, <https://doi.org/10.1080/15376494.2018.1444235>.
 - [57] Y.H. Dong, L.W. He, L. Wang, Y.H. Li, J. Yang, Buckling of spinning functionally graded graphene reinforced porous nanocomposite cylindrical shells: an analytical study, *Aerosp. Sci. Technol.* 82–83 (2018) 466–478, <https://doi.org/10.1016/j.ast.2018.09.037>.
 - [58] O. Polit, C. Anant, B. Anirudh, M. Ganapathi, Functionally graded graphene reinforced porous nanocomposite curved beams: bending and elastic stability using a higher-order model with thickness stretch effect, *Composites, Part B, Eng.* 166 (2019) 310–327, <https://doi.org/10.1016/j.compositesb.2018.11.074>.
 - [59] M.R. Barati, A.M. Zenkour, Analysis of postbuckling of graded porous GPL-reinforced beams with geometrical imperfection, *Mech. Adv. Mat. Struct.* 26 (2019) 503–511, <https://doi.org/10.1080/15376494.2017.1400622>.
 - [60] M. Sobhy, A.M. Zenkour, Wave propagation in magneto-porosity FG bi-layer nanoplates based on a novel quasi-3D refined plate theory, *Waves Random Complex Media* 0 (2019) 1–21, <https://doi.org/10.1080/17455030.2019.1634853>.
 - [61] D.X. Hung, T.M. Tu, N. Van Long, P.H. Anh, Nonlinear buckling and postbuckling of FG porous variable thickness toroidal shell segments surrounded by elastic foundation subjected to compressive loads, *Aerosp. Sci. Technol.* 107 (2020) 106253, <https://doi.org/10.1016/j.ast.2020.106253>.

- [62] H. Babaei, M. Reza Eslami, Size-dependent vibrations of thermally pre/post-buckled FG porous micro-tubes based on modified couple stress theory, *Int. J. Mech. Sci.* 180 (2020) 105694, <https://doi.org/10.1016/j.ijmecsci.2020.105694>.
- [63] M. Mohammadi, M. Bamdad, K. Alambeigi, R. Dimitri, F. Tornabene, Electro-elastic response of cylindrical sandwich pressure vessels with porous core and piezoelectric face-sheets, *Compos. Struct.* 225 (2019) 111119, <https://doi.org/10.1016/j.compstruct.2019.111119>.
- [64] H.S. Shen, F. Lin, Y. Xiang, Nonlinear bending and thermal postbuckling of functionally graded graphene-reinforced composite laminated beams resting on elastic foundations, *Eng. Struct.* 140 (2017) 89–97, <https://doi.org/10.1016/j.engstruct.2017.02.069>.
- [65] W.J. Craft, R.M. Christensen, Coefficient of thermal expansion for composites with randomly oriented fibers, *J. Compos. Mater.* 15 (1981) 2–20, <https://doi.org/10.1177/002199838101500102>.
- [66] R.B. Hetnarski, M.R. Eslami, *Thermal Stresses—Advanced Theory and Applications*, Springer, The Netherlands, 2009.